

Lets talk Open Source

The power and utility of mobile devices is always on the rise—cell phones that double up as PDAs, PDAs that can do multimedia and so on. This scenario is about to get better with the cell phone bigwigs showing a trend of adopting open source technologies, paving the way for third-party developers to create applications that can talk to the next generation of mobile



devices. For example, the new Nokia 7650 phone has aborted the Psion Link Protocol (PLP) that was used by Symbian and

has now moved over to the standard TCP/IP protocols. Work on this is being carried out by Jeremy Burton of a company called Intuwave, in conjunction with Mark Melling who earlier worked with Symbian.

With the path now paved for developing on open protocols, the route is far easier for Linux and Macintosh developers to create software for these new phones. Already Linux and KDE hackers have successfully integrated support for Psion-based handheld computers directly into the operating system's interface.

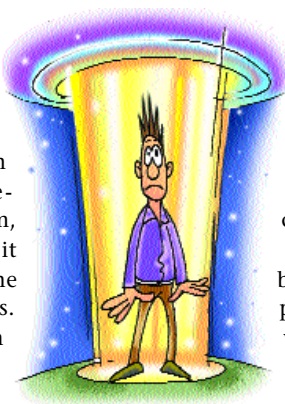
With many such advances on the way, the erstwhile PLP protocol can now be safely relegated to the recycle bin since Psion does not make PDAs any more and the protocol is quite dated.

Aussies teleport laser beam

A team of physicists at the Australian National University led by Dr Ping Koy Lam has successfully disembodied a laser

beam at one location and rebuilt it at another spot about a meter away and in the blink of an eye. The team teleported the beam, *Star Trek* style, but it got destroyed in the recreation process. However, Lam claims that day is not too far when solid matter could

be teleported from one location to another. "My prediction is that it will be done in the next three to five years—that is the teleportation of a single atom." But since humans are made up of zillions of atoms (one followed



by 27 zeroes), the possibility of a *Star Trek* like transporter is remote. "In theory, there is nothing stopping us from

doing it, but the complexity of the problem is so huge that no one is thinking seriously about it at the moment," Lam told a news conference.

The immediate benefits of such a process lie elsewhere. The breakthrough opens up enormous possibilities for super-fast and super-secure communications systems and quantum computing, over the next decade. Physicists believe quantum computers have the ability to solve problems millions of times faster than conventional computers.

The new Celeron

Intel unveiled its first Celeron processor based upon a 0.18 micron architecture with a Pentium 4-derived core.

Debuting at 1.7 GHz, this processor will feature 128 KB of L2 cache as opposed to the 256 KB of cache found in the P4 processor. Architecturally, the core of this new Celeron is no different from the Willamette core of the P4 that was launched in November 2000.

This processor operates on motherboards based on Intel's i845 and the i850 chipsets. Due to DDR and RDRAM support offered by

these chipsets, their memory bandwidth would be significantly higher than that offered by the erstwhile i815 motherboards.

The new Celeron uses the same Quad-pumped bus that is utilised by the P4, which translates into 3.2 GBps of bandwidth—enough to support this processor at its current speed.

The icing on the cake? It's priced nearly 40 per cent cheaper than the P4 1.7 GHz. Tests have also indicated that this processor lends itself well to overclocking and could become a poor man's P4 if tweaked!



C for graphics

Programmable video cards have been with us for some time now, but the promise of special effects and real-time cinema-quality graphics remains largely unfulfilled. A large part of the blame can be put squarely on the need for assembly-level programming to get anything good out of these video cards.

Working at a hardware level is a time-consuming and non-intuitive process that few programmers wish to get into. This is where nVidia hopes its latest offering—a high-level language termed Cg, 'C for Graphics'—will offer an easier solution.

Cg is an attempt to replace OpenGL and Direct3D and also offers cross-API transparency in which the same code can run on Windows, Linux, Mac OS X and the Xbox. Cg will work with both, nVidia and non-nVidia hardware and is compatible with the upcoming DirectX 9.0 High Level Shading Language. In short, Cg will allow game developers to utilise powerful pixel and vertex shader effects and finally put them to their intended use.

As it approaches maturity (expected by October this year) a better picture will emerge on its capabilities and limitations.

snapshot

**VSNL stands to lose
Rs 3.5 billion
if WorldCom Inc
files for bankruptcy
following the
accounting scandal**

Source: Reuters

■ Pac-Man headed for Sprint 3G ■ RealNetworks to launch GamePass monthly subscription to give free and discounted games ■ Doom III possible for Xbox